

# Estimating Volumes of Square and Rectangular Pyramids

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# Recall

The volume of a pyramid is about **one-third** the volume of a prism with the same base and height.

$$V = \frac{1}{3}s^2h \quad (\text{square})$$

$$V = \frac{1}{3}lwh \quad (\text{rectangular})$$

To estimate the volume, round the dimensions to convenient numbers before computing.

# Why Estimate?

Estimation is useful when exact measurements are unnecessary or when making quick calculations.

# Rounding to Convenient Numbers



8.8 is closer to 9, so  $8.8 \approx 9$ .

# Helpful Tip

The formula has  $\frac{1}{3}$  in it.

If possible, round to a number that is easy to divide by 3, such as 3, 6, 9, 12, or 15.

# Steps to Estimate the Volume

1. Round each dimension to a convenient number.
2. Substitute into the volume formula.
3. Multiply, then divide by 3.
4. Write the answer using  $\approx$  and **cubic units**.

Memory Tip

**Round, Multiply, Divide by  
3**

# Reasonable Estimates

Compare the estimated answer with the exact value to determine if the estimate is reasonable.

A good estimate is close to the exact value.

# Examples

# Example 1

Estimate the volume of a square pyramid with a base edge of 6.1 cm and a height of 8.8 cm.

$$6.1 \approx 6 \qquad 8.8 \approx 9$$

$$\begin{aligned} V &= \frac{1}{3}s^2h \\ &\approx \frac{1}{3}(6)^2(9) \\ &\approx \frac{1}{3}(324) \\ &\approx 108 \text{ cm}^3 \end{aligned}$$

# Example 2

Estimate the volume of a rectangular pyramid with a base measuring 10.2 cm by 5.8 cm and a height of 8.7 cm.

$$10.2 \approx 10 \quad 5.8 \approx 6 \quad 8.7 \approx 9$$

$$\begin{aligned} V &= \frac{1}{3}lwh \\ &\approx \frac{1}{3}(10)(6)(9) \\ &\approx \frac{1}{3}(540) \\ &\approx 180 \text{ cm}^3 \end{aligned}$$

# Example 3

Estimate the volume of a square pyramid with a base edge of 14.9 m and a height of 11.2 m.

$$14.9 \approx 15 \qquad 11.2 \approx 11$$

$$\begin{aligned} V &= \frac{1}{3}s^2h \\ &\approx \frac{1}{3}(15)^2(11) \\ &\approx \frac{1}{3}(2\,475) \\ &\approx 825 \text{ m}^3 \end{aligned}$$

# Example 4

Estimate the volume of a rectangular pyramid with a base of 20.3 m by 14.8 m and a height of 8.9 m.

$$20.3 \approx 20 \quad 14.8 \approx 15 \quad 8.9 \approx 9$$

$$\begin{aligned} V &= \frac{1}{3}lwh \\ &\approx \frac{1}{3}(20)(15)(9) \\ &\approx \frac{1}{3}(2\,700) \\ &\approx 900 \text{ m}^3 \end{aligned}$$

# Example 5

Is the estimate in Example 1 reasonable?

$$\text{Estimate} \approx 108 \text{ cm}^3$$

$$\text{Exact} = \frac{1}{3}(6.1)^2(8.8) \approx 109.15 \text{ cm}^3$$

The two values are close, so the estimate is reasonable.

# Practice Exercises

# Practice 1

Estimate the volume of the pyramid.

A square pyramid has a base edge of 8.2 cm and a height of 11.6 cm.

## Practice 2

Estimate the volume of the pyramid.

A square pyramid with a base edge of 12.4 m and a height of 9.7 m.

## Practice 3

Estimate the volume of the pyramid.

A rectangular pyramid with a base of 15.3 m by 10.6 m and a height of 12.2 m.

## Practice 4

Estimate the volume of the pyramid.

A square pyramid has a base edge of 19.6 cm and a height of 14.8 cm.

## Practice 5

Estimate the volume of the pyramid.

A rectangular pyramid with dimensions 13.5 cm, 8.2 cm, and height 16.9 cm.

## Practice 6

Estimate the volume of the pyramid.

A square pyramid has a base edge of 24.7 m and a height of 18.3 m.

## Practice 7

Estimate the volume of the pyramid.

A rectangular pyramid whose base is 18.8 cm by 11.4 cm and height is 15.6 cm.

## Practice 8

Estimate the volume of the pyramid.

A square pyramid has a base edge of 31.2 cm and a height of 20.5 cm.

# Activities

# Activity 1

Estimate the volume of the pyramid.

A square pyramid has a base edge of 9.4 cm and a height of 12.3 cm.

## Activity 2

Estimate the volume of the pyramid.

A square pyramid with a base edge of 13.8 m and a height of 10.5 m.

## Activity 3

Estimate the volume of the pyramid.

A rectangular pyramid has a base measuring 8.6 cm by 6.4 cm and a height of 11.8 cm.

## Activity 4

Estimate the volume of the pyramid.

A rectangular pyramid with a base of 16.9 m by 11.5 m and a height of 13.6 m.

## Activity 5

Estimate the volume of the pyramid.

A square pyramid has a base edge of 21.1 cm and a height of 15.9 cm.

## Activity 6

Estimate the volume of the pyramid.

A rectangular pyramid with dimensions 14.8 cm, 9.7 cm, and height 17.4 cm.

## Activity 7

Estimate the volume of the pyramid.

A square pyramid has a base edge of 26.3 m and a height of 19.5 m.

## Activity 8

Estimate the volume of the pyramid.

A rectangular pyramid whose base is 20.4 cm by 12.8 cm and height is 16.7 cm.

## Activity 9

Estimate the volume of the pyramid.

A square pyramid has a base edge of 32.8 cm and a height of 21.9 cm.

## Activity 10

Estimate the volume of the pyramid.

A rectangular pyramid with a base measuring 24.3 m by 18.6 m and a height of 26.8 m.

Thank you!